



# Vojtěch Špaček

**Senior C++ engineer — safety-critical embedded & HIL test automation | Linux/networking | agentic-AI (MCP) tooling**

*Pragmatic programmer and creative problem solver — systems, verification and open source.*

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**Nationality:** EU (Czech Republic) **Place of residence:** Brno, CZ

**Work permit:** EU citizen (Czech Republic) — no permit required in the EU/EEA. B-permit eligible (EU citizen).

## Profile Summary

Senior C++ engineer working on safety-critical and real-time systems: aircraft onboard diagnostics at Honeywell, electron-microscopy detector software at Thermo Fisher Scientific, and a Linux inspection proxy and firewall at Trusted Network Solutions.

Most recently sole owner of the hardware-in-the-loop verification suite and automated bench for a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), raising measured conformance to 259 of 271 tracked state transitions (95.6%).

Linux, networking and security specialist — Netlink, nftables, TLS inspection, Kerberos/LDAP/OCSP — with C++20 daemons shipped as Debian packages and a Rust reimplementation published on crates.io.

Builds agentic-AI tooling on the Model Context Protocol: an agent harness adopted into a daily client delivery workflow, and MCP-Prompts, published to npm, Docker Hub and the MCP directories (Glama, MCP Market, PulseMCP) with 97 stars and 21 forks.

## Work Experience

### Test Automation Engineer (Python, C#)

Resideo (Ademco) (via CoolPeople (contractor))

Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), as sole owner of the verification test suite and of the automated bench that drives it. Owned delivery end to end: requirement traceability, issue tracking, branch and pull-request workflow, Conan packaging, CI integration and monthly client reporting.

- Designed an Arrange/Act/Assert validation pipeline in Python with eight independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.
- Built a data-driven generator expanding a YAML catalogue of 270+ state transitions, 38 states and 33 events into roughly 290 executable tests, removing hand-written duplication.
- Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering a pyvisa/USBTMC/SCPI programmable AC source, CAN serial-bus monitoring and Modbus relay and probe I/O.

### Software Developer (C++)

Thermo Fisher Scientific (via Ixperta (contractor)) · 2023 – 2024

Electron microscopy software for high-resolution detector hardware operation and image acquisition. Built server-side C++ applications, FPGA-based data processing pipelines with gRPC networking, and backend logic for a C#/NET WPF image processing and display application. Worked in an international Agile team with bi-weekly stakeholder demos.

- Implemented server-side software in C++, including hardware abstraction layers (HAL) and inter-process communication (IPC) using COM and MIDL on Windows.
- Developed embedded software for FPGA hardware, using gRPC network communication in C++, Python and Linux environments.
- Optimized the data acquisition pipeline and implemented robust error handling.

### Software Engineer II

Honeywell · 2022 – 2023

Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight. Architected and implemented the DiagTest framework, a state machine-based diagnostic test engine in C++ managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

- Designed a job-based execution system with DiagTestJob and DiagTestMsgQueueProc for asynchronous test orchestration, implementing message queue processing patterns for reliable real-time diagnostics.
- Engineered a test interference management system preventing conflicting diagnostic tests from running simultaneously, ensuring aircraft system safety and data integrity.
- Implemented four diagnostic test modes (data collection, passive monitoring, interactive, non-interactive) with timeout management, precondition handling and aviation-standard command/response protocols.

### Application Developer

Trusted Network Solutions · 2020 – 2022

Kernun Adaptive Firewall. Implemented a forward proxy in C++ on Linux featuring HTTP/HTTPS proxying with TLS/SSL inspection, OCSP integration for certificate validation, Kerberos authentication and LDAP user/group lookup, plus the ipmon network monitoring daemon driving nftables firewall rules from real-time IPv4/IPv6 address changes.

- Built comprehensive proxy infrastructure with certificate generation and management (CA, client certificates), a dynamic configuration builder/parser system, and a manager/partition architecture for modular daemon design.
- Integrated security components including Suricata (intrusion detection) and nftables (static firewall), aggregating logs and managing connection information in PostgreSQL.
- Implemented real-time network interface monitoring in ipmon using NETLINK sockets, handling both IPv4 and IPv6 address management, with Unix domain sockets for IPC and dynamic nftables rule management.

## Independent IT Contractor

Self-employed

Independent contracting for business and consumer clients: CI/CD automation pipelines and Linux server administration for B2B customers, and IT support, consumer hardware retail and security consulting for B2C customers.

- B2B: CI/CD automation pipelines and Linux server administration.
- B2C: IT support freelancing, consumer hardware retail and security consulting.

## Automation Developer & IT Systems Specialist

IBM · 2014 – 2016

Enterprise IT operations: infrastructure monitoring, incident management and an automation solution that replaced manual ticketing and server administration work.

- Developed an automation application that replaced manual workload in ticketing operations and basic server administration, using Bash, PowerShell and JavaScript for cross-platform server management and applying automata theory to design state-driven workflows.
- Monitored Unix/Windows servers and resolved production incidents, working hands-on with Active Directory and VMware virtualization.

## Skills

**Embedded & RTOS:** C++ (C++17/C++20) (Senior) · Embedded software / ARM / RTOS (Medior) · State machine design patterns (Senior) · Hardware abstraction layers & IPC (COM/MIDL, Unix sockets) (Medior) · FPGA-adjacent development & gRPC (Medior) · Flatbuffers (binary serialization) (Medior) · Qt/QML (Working knowledge)

**Verification & HIL:** Python (Medior) · Hardware-in-the-loop test design (Senior) · pyvisa / USBTMC / SCPI instrument control (Medior) · Modbus I/O cards & CAN serial-bus diagnostics (Medior) · Safety-certified product verification (UL/CSA, ANSI Z21.20, NFPA 54) (Medior) · Requirement traceability & test evidence reporting (openpyxl, HTML, Mermaid) (Medior) · C# / .NET (Medior) · Perl test frameworks (Medior)

**Networking & Security:** Linux (Senior) · Linux networking, TCP/IP, Netlink, nftables (Senior) · Encrypted traffic inspection (TLS 1.3, OCSP) (Medior) · Kerberos/Heimdal, LDAP, PostgreSQL, Suricata IDS (Medior) · Forward/TCP/HTTP proxy design (Senior) · Rust (Working knowledge)

**AI / LLM / MCP:** Model Context Protocol server development (Senior) · Agentic workflow design & governance (agents, skills, policy hooks) (Medior) · Prompt engineering (RAG, few-shot, prompt caching, structured outputs, tool calling) (Medior) · LLM provider APIs & SDK integration (Medior) · JavaScript / TypeScript (Medior) · Node.js (Medior)

**Tooling:** Git (Senior) · Bash (Senior) · Docker / Podman (Medior) · Conan (C/C++ dependency management) (Medior) · CI/CD (Jenkins, TeamCity, GitLab CI, GitHub Actions) (Medior) · PowerShell (Medior) · Enterprise Architect (UML) (Working knowledge) · GDB, Wireshark, VirtualBox (Medior)

## Open Source & Projects

**MCP-Prompts** — 97 stars, 21 forks | [github.com/sparesparrow/mcp-prompts](https://github.com/sparesparrow/mcp-prompts)

TypeScript · Node.js · PostgreSQL + pgvector · Docker · REST + SSE · AWS

**rust-network-mgr** | [github.com/sparesparrow/rust-network-mgr](https://github.com/sparesparrow/rust-network-mgr)

Rust · Netlink · nftables · epoll · Lock-free data structures

**rtp-midi** — 2 stars | [github.com/sparesparrow/rtp-midi](https://github.com/sparesparrow/rtp-midi)

Rust (66.5%) · RTP-MIDI · OSC · Android NDK (Kotlin) · ESP32 / FreeRTOS · Docker

## Education & Training

**Masaryk University, Faculty of Informatics** — Computer Science

*Coursework only — no academic title* — Coursework in programming, Linux and networking.

**Coursera.org / DeepLearning.AI** — Machine learning and LLM engineering

*Coursework only — no academic title* — Deep Learning Specialization (four machine learning courses by Andrew Ng), plus data analysis, prompt engineering and building LLM agents.

**Self-directed study** — Design patterns, software architecture and offensive security

*Coursework only — no academic title* — Design Patterns (refactoring.guru) and Hack The Box cybersecurity training; self-studied the Certified Ethical Hacker syllabus (exam not taken).

**Professional development reading** — Software engineering practice

*Coursework only — no academic title* — Software Engineering at Google, The Pragmatic Programmer, Clean Architecture.

## Languages

Czech (native) · Slovak (native) · English (C1) · German (A1)

## References

References available on request — two to three referees and work certificates (Arbeitszeugnisse) can be provided on request.

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